

Digital Image Processing & Principles of Management

Mid-Term Examination

Total Marks: 20

Time Allowed: 90 minutes

Section A - Multiple Choice Questions (MCQs)

Instructions: Each question carries 1 mark. Select the most appropriate option (A, B, C, or D).

Q1. The Lempel Ziv (LZ) coding algorithm achieves compression primarily by:

- A) Reducing the resolution of the image
- B) Building a dictionary of previously seen strings
- C) Converting pixels into frequency domains
- D) Using fixed-length codes for all characters

Q2. In the Lempel Ziv algorithm, how is a newly identified phrase coded?

- A) By its frequency of occurrence
- B) By its ASCII value
- C) By the location of its prefix and the value of the last bit
- D) By the total length of the input sequence

Q3. Human Resource Management (HRM) is defined as a set of organizational activities directed at:

- A) Financial auditing and tax reporting
- B) Attracting, developing, and maintaining an effective workforce
- C) Managing the physical supply chain and logistics
- D) Programming and software development

Q4. Which concept reflects an organization's investment in attracting, retaining, and motivating an effective workforce?

- A) Physical Capital
- B) Working Capital
- C) Human Capital
- D) Structural Capital

Q5. According to the Lempel Ziv algorithm, the source sequence is parsed into:

- A) Fixed blocks of 8 bits
- B) Strings that have not appeared so far

C) Mathematical equations

) Random segments of equal length

Q6. Internal recruiting is best described as:

A) Locating applicants through social media

B) Hiring candidates from competing firms

C) Considering present employees for promotions or transfers

D) Attracting fresh graduates from universities

Q7. Which of the following is an "External Condition" imposed on a group that affects its performance?

A) Member personality traits

B) Formal rules and regulations

C) Interpersonal conflict management

D) Individual member skills

Q8. In the context of group performance, which skills are specifically mentioned as important for high performance and collaborative problem solving?

A) Technical coding skills

) Interpersonal skills

C) Manual labor skills

D) Financial accounting skills

Q9. When using LZ coding, if a sequence is "too random," the resulting compressed sequence may be:

A) Much shorter than the original

B) Exactly half the size

C) Longer than the original sequence

D) Automatically deleted

Q10. The process of locating, identifying, and attracting capable applicants to an organization is known as:

A) Selection

B) Recruiting

C) Orientation

D) Job Analysis

Section B - Theory / Descriptive Questions

Instructions: Answer the following questions in the space provided. Each question carries 5 marks.

Q1. [Bloom's Level: L2 - Understand]

Describe how the Lempel Ziv (LZ) algorithm sequentially parses a source sequence and what determines the shortest string to be marked off.

Q2. [Bloom's Level: L1 - Remember]

Identify the specific organizational resources and factors that constitute the external conditions affecting group performance and satisfaction.